A SAI MOULI

Backend Software Engineer

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PROFESSIONAL SUMMARY

Backend engineer at NatWest Group, building backend APIs and a natural-language-to-SQL service over a 136-table reporting database. Builds multi-tenant FastAPI/PostgreSQL systems with policy-gated agent workflows; merged fixes into FastMCP and DSPy.

EXPERIENCE

Software Engineer | NatWest Group Bengaluru, India | Jul 2025 – Present

- Cut large-query dashboard response time from **~3 min to under 30 s** by reworking the backend APIs and data path.
- Built a natural-language-to-SQL engine over a **136-table** reporting database: it turns a plain-English question into SQL, validates the query against the schema before it runs, executes it, and returns the answer with a chart.
- Shipped the engine as a chatbot in the analytics dashboard, so stakeholders pull ad-hoc reports themselves.

Backend Developer Intern | Aamara Technologies Bengaluru, India | Mar 2024 – Aug 2024

- Built the backend of Techno Rings' shop-floor operations system, used by **20,000 people**, on Django REST Framework and MySQL with a React front end: machine assignment, shift logging and tool-life tracking.
- Implemented payroll exception paths as model-level rules: breakdowns pro-rate shift and target, and workers move to another machine's job. Daily processing fell from **about four hours to under thirty minutes**.

Software Development Intern | Pods Studio Bengaluru, India | Nov 2023 – Dec 2023

- Moved customer email outreach onto event-driven Celery tasks and cron jobs, removing the manual send step (retention up **27%**); root-caused and fixed customer-reported failed in-app payments.

OPEN-SOURCE CONTRIBUTIONS

PrefectHQ/fastmcp 2 PRs merged

- **PR #4758 (merged):** fixed a lost-update race in the event store (a five-concurrent-write reproduction kept 1 of 5 event IDs); after maintainer review, replaced a store-wide lock with **64 striped asyncio locks** keyed by stream ID.
- **PR #4705 (merged):** stopped the response cache storing is_error results, which pinned a transient 503 for a full hour.

stanfordnlp/dspy | **PR #10720 (merged):** fixed dspy.majority() raising TypeError when voting on non-string values 1 PR merged

chroma-core/chroma 1 issue, 1 open PR

- **Issue #7560:** showed the CI lint job could never fail (continue-on-error plus bash -e meant black, flake8 and prettier never ran). **PR #7559 (open)** proposes a fix for two undefined-name defects it hid.

567-labs/instructor | **PR #2846 (open):** renders context placeholders in input_text parts that reached the model raw 1 open PR

PROJECTS

Zippyy | Python, FastAPI, PostgreSQL, Redis, Next.js github.com/SaiMouli3/zippyy

- Built a multi-tenant delivery-recovery platform: signed carrier, WhatsApp and voice webhooks are de-duplicated into an inbox, drive a shipment state machine, and wake durable agent workflows that act only through typed tools.
- Every carrier action must pass a deterministic 14-rule policy engine, and the LLM reads only replies the rule extractor is unsure of. Per-merchant row-level security, SKIP LOCKED job leases; 145 tests pass.

XTXT | Go, Python, JavaScript, Rust, Java, C github.com/SaiMouli3/xtxt

- Designed a plain-text document format where structure is explicit syntax: directives carry images, tables, charts and typed records that the CLI extracts as JSON, while files still diff cleanly in Git.
- Wrote the spec, a Go reference parser and CLI, and a 12-case conformance suite the Python, JS, Rust, Java and C SDKs run in CI; published to PyPI and npm, with a VS Code extension and MCP server.

TECHNICAL SKILLS

Languages: Python, C++, SQL
Backend Engineering: FastAPI, Django, Django REST Framework, Flask, SQLAlchemy, REST APIs, WebSockets, Celery
AI Engineering: LangGraph, LangChain, MCP servers, RAG pipelines, embeddings, FAISS
Databases & Infrastructure: PostgreSQL, MySQL, Redis, AWS, Docker, Kubernetes, Linux, Git, GitHub Actions
Engineering Practices: asyncio concurrency, database transactions, pytest, API design, performance optimization

EDUCATION

Ramaiah Institute of Technology | B.Tech, Computer Science and Engineering | CGPA: 9.06/10 2021 – 2025